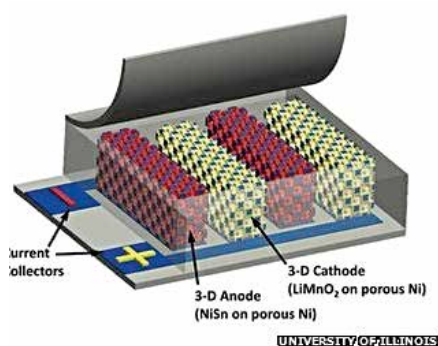


than current batteries. The 3D electrodes have small intertwined fingers that reach into each other thereby increasing the surface area and bringing the anode and cathode closer, so that the ions and electrons do not have to travel far. The principle is the same; electrons moving from the anode to the cathode when the battery is under use and charged ions moving via the electrolyte. The researchers have found a way to integrate the anode and cathode at a microscale. The transfer of electrons in the 3D microbattery array is happening at a micro-scale level, and the distance between cathode



3D microbatteries

and anode is less, the energy generated is quite high. As the electrodes are closer, even recharging is speedy. The best part is that the researchers claim that their technology is scalable. So you can have microbatteries powering your cellphones as well as your electric car. Also the size of the batteries will be much smaller. The only concern at the moment is with the issue of robustness – any short circuit within the microbattery can be hazardous. This sentiment is echoed by the head of the research group Professor King, who states that the current microbatteries use a combustible electrolyte. The aim is to move to non-combustible polymer electrolytes. According to Prof. King, the trials of his microbattery as a power source in electronic devices should begin by the end of year.

Supergirl's supercapacitor

While our phone and tablets are increasing their battery capacities, the



Eesha Khare with her supercapacitor prototype

amount of time taken to charge these devices is also increasing progressively. How about reducing that time to under a minute?

Well, an 18-year old Indian-American Eesha Khare has developed a supercapacitor that she claims can charge a cellphone in 20 seconds. She has developed a supercapacitor that can hold a large amount of energy in a small volume. It is quite small and can easily fit inside cellphone batteries and other electronic devices, thereby allowing them to fully charge between 20-30 seconds. Another impressive feat of this supercapacitor is its ability to undergo upto 10,000 charge-recharge cycles which is 10 times the number of cycles current cellphone batteries are capable of. Also Eesha's supercapacitor is flexible which immediately makes it eligible for use in the future in rollable displays and even electric car batteries. Although she has only demoed this tech using an LED bulb so far, this invention is so impressive that it won Eesha the Intel Foundation Young Scientist Award this year from among 1,700 participants from 70 countries.

The implications of this invention are pertinent to each and everyone of us, as we are tired of long charging times. We just hope this technology sees the light of the day pretty soon. Fingers crossed.

Graphene supercapacitors

Many scientific inventions and discoveries can be attributed to accidents – the scientist tending to stumble upon one thing while looking for something else. Something similar happened with researcher Richard Kaner and his student Maher El-Kady of the California

NanoSystems Institute at UCLA. After devising a way to make super-thin (one-atom thick) graphene wafer sheets using a DVD burner of all devices, El-Kady decided to try lighting a bulb with one of the small square of those graphene sheets – and it worked!

That was the Eureka moment! When the researchers realised that they had stumbled upon an energy storage medium which also has the ability to charge/discharge quickly and we are talking in seconds here. Generally, the batteries that we use are high energy storage devices but have painfully slow charge/discharge cycles whereas a capacitor has a low energy, high power and very quick charge/discharge times. The graphene micro-supercapacitor which is what the researchers are working on is both high energy and high power device with quick charge/recharge cycles.

Traditionally fabricating graphene micro-supercapacitor involved a labour-intensive process, but Kaner and El-Mady's method of using a DVD burner to make thin graphene sheets if scaled can work out to be a much cheaper way to produce the micro-supercapacitors.

You have over 100 micro-supercapacitors on a single disc in under 30 minutes. For any supercapacitor to be effective, the two separate electrodes have to be positioned in a way that the surface area between them is maximised so as to hold more charge.

The researchers from UCLA placed the electrodes side-by-side using an interdigitated pattern thereby increasing surface area available for each of the electrodes at the same time reducing the distance for the ions to travel (something similar to what we saw in microbatteries above). Since these graphene supercapacitors are flexible, bendable, twistable in nature they can be used as energy-storage devices in flexible electronics such as roll-up display or TVs, epaper as well as wearable electronics. Currently the team is looking for industry partners to help them mass produce the graphene micro supercapacitors. Video: <http://digit.in/16fG8ii>.